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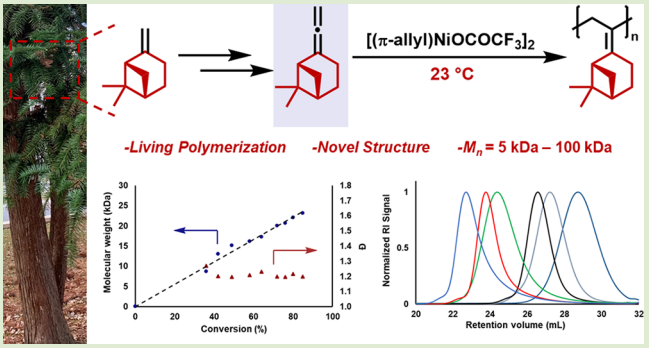
Controlled Polymerization of β-Pinadiene: Accessing Unusual Polymer Architectures with Biomass-Derived Monomers

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ABSTRACT: Biomass-derived polymers are emerging as critically needed alternatives to their petrochemical counterparts. Terpenes, which are among the most abundant natural products, represent particularly fertile chemical space for monomer development (given their inherent structural complexity). Here, we present the living vinyl-addition polymerization of β-pinadiene (the cumulated congener of β-pinene) at room temperature. Employing [(π-allyl)NiOCOCF$_3$]$_2$ as a catalyst afforded the desired polymers with good control over molecular weight and dispersity. Interestingly, the bicyclic pinane core was retained in the isolated materials (which starkly contrasts prototypical pinene polymerizations). Moreover, the reported materials exhibited impressive thermal stability (T_d = 294 °C) and high glass transition temperatures (T_g = 160 °C). As the polymerization of terpene-derived cumulenes can afford scaffolds that defy current synthetic logic, we anticipate our work will unlock additional avenues for sustainable polymer development.



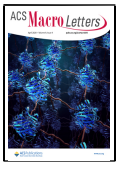
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As the demand for commodity plastics has increased, so too has global production of synthetic polymers.[1] Nearly 350 Mt of these materials are generated every year,[2] and over 40% of these polymers are polyolefins.[2] As polyolefins are predominantly derived from petrochemical feedstocks, their preparation on such a scale necessitates significant oil consumption. Indeed, nearly 10% of crude oil production is diverted to polymer manufacturing (a figure that is expected to double by 2050).[3] The rising demand for polyolefins starkly contrasts the availability of the raw materials for their production; thus, the development of more sustainable alternatives is imperative. Biological feedstocks offer a rich diversity of potential monomers (both in terms of chemical functionality and structural complexity), and they are increasingly becoming focal points for advancements in polymer science. For example, polylactide is a commercially viable alternative to polyethylene and polystyrene in certain applications.[4,5] Biobased diamines and dicarboxylic acids have also emerged as precursors for polyamides used in the automotive[6] and medical industries.[7] As such, the investigation of novel, biologically derived monomers is essential for realizing next-generation renewable materials.

Terpenes, which are common plant oils, have been investigated as monomers since the early 20th century,[8] and synthetic polyterpenes have found a myriad of uses (e.g., as tackifiers,[9] adhesives,[10] food packaging materials,[11] and latex substitutes[12]). Pinenes are particularly promising candidates for renewable monomers, given their high natural abundance.[13] While both α- and β-pinene undergo polymerization, β-pinene reacts more readily due to facile ring opening (and concomitant relief of ring strain).[14]

While cationic polymerization is the most reliable method for preparing poly(β-pinene),[15] this approach suffers from several limitations. For example, high catalyst loadings and incremental monomer additions are common requisites. Cryogenic conditions are also typical, although Lu and co-workers recently disclosed the polymerization of β-pinene to modest number-average molecular weight (M_n = 10 900 Da) at elevated temperatures.[16] Moreover, the preparation of polymers with high molecular weight (M_n > 50 000 Da) has proven challenging. In fact, the first synthesis of a relatively high M_n (cf. 50 000 Da) pinene polymer was only recently disclosed by Kamigaito et al.[17] The team reported a living polymerization mediated by EtAlCl$_2$, but incremental monomer addition and cryogenic temperatures were required to achieve good conversion. Additionally, the catalyst exhibited apparent deactivation at longer reaction times. As such, opportunities remain for improving access to pinene-derived polymers.

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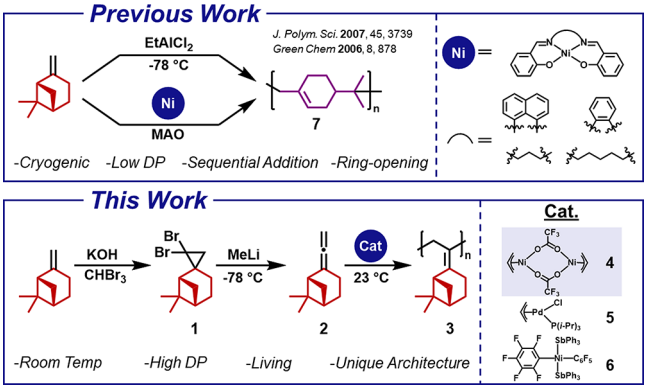
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We became intrigued by β-pinene as a monomer not only because of its relative abundance but also because of the bicyclic nature of the pinane core. The incorporation of rigid carbocycles into polymers can dramatically alter bulk properties,[18−22] but there have been no reports detailing materials that contain a pinane motif. As traditional polymerizations of β-pinene proceed with concomitant destruction of the pinane bicycle,[14,17,23,24] we sought to develop an alternative strategy that would circumvent this reactivity. Specifically, we hypothesized that retention of the bicyclic system would be possible if propagating intermediates were isolated on exocyclic positions.

Extending the olefin through cumulation could potentially enable such a scenario, especially if the cumulene exhibited selective 2,3-polymerization tendencies. Allenes are known to polymerize under a variety of conditions (e.g., radical[25] or coordination−insertion[26−28]), but these approaches typically exhibit poor regioselectivity. Endo, however, has demonstrated that hindered, 1,1-disubstituted allenes undergo living polymerizations with excellent selectivity for 2,3-insertions in the presence of a (π-allyl)Ni catalyst (**4**, Scheme 1).[29] We

Scheme 1. Synthetic Strategies to Access Pinene-Derived Polymers



envisioned that the pinane core would be sufficiently encumbering to promote exclusive 2,3-polymerization under the action of Endo's catalyst. Furthermore, we hypothesized that selective 2,3-polymerization of β-pinadiene would preclude ring opening; thus, we could potentially interrogate polymer properties associated with this repeating unit.

Our investigation began with the synthesis of β-pinadiene monomer, **2**. Treatment of (1*S*)-(−)-β-pinene with bromoform and KOH under phase transfer conditions smoothly afforded **1** in excellent yield (93%; Scheme 1; see the Supporting Information for additional details). We found this reaction could be conducted on a decagram scale without the need for tedious purification strategies. Under the action of MeLi, **1** undergoes a Skattebøl rearrangement[30] to afford **2**, which could be isolated in good yield (85%) following distillation. Moreover, neat samples of **2** could be stored indefinitely at −20 °C.

Given the enhanced regioselectivity reported by Endo,[29] we elected to use vinyl-addition catalyst **4** in an initial polymerization attempt (Table 1; entry 1). Quantitative consumption of **2** in toluene was observed after 70 min (as determined by ^{1}H NMR spectroscopy), at which point the reaction mixture was added to MeOH to precipitate any polymeric material. Filtration afforded **3** as a white powder in excellent yield (93%). The polymer was analyzed by gel permeation chromatography (GPC), which revealed an M_n value in good agreement with the theoretical value (cf. 39 210 Da versus 30 000 Da) and narrow dispersity ($Đ$ = 1.16). Although we surveyed other common vinyl addition catalysts (precatalyst **5**, and **6**; Scheme 1; SI), neither outperformed **4** (Table 1).

^{1}H NMR spectroscopic analysis of the material prepared using catalyst **4** was consistent with the expected 2,3-regioselectivity, given the absence of observable vinyl resonances (Figure 1). Importantly, however, these data also suggested the retention of the bicyclic pinane core. Indeed, ring opening of this motif would be expected to afford a new vinylic signal from the associated endocyclic olefin (*vide infra*). To further investigate the material's structure, we performed a series of correlational and heteronuclear single quantum coherence (HSQC) ^{13}C−^{1}H NMR spectroscopy experiments. To our delight, these data confirmed the ^{13}C resonances associated with the pinane bridgehead methines (cf. 46.4 and 40.7 ppm) and methylenes (cf. 28.2, 24.7, and 21.8 ppm) were retained in the polymer (Figure 1; Figures S19, S20, S23, and S24).

Having established the superior reactivity of **4**, we sought to further optimize the polymerization conditions. **4** performed poorly in acetonitrile, affording only 14% conversion of **2** after 70 min (as determined by ^{1}H NMR spectroscopy; Table 1). Tetrahydrofuran improved monomer conversion, but the isolated polymer exhibited broad dispersity and a calculated M_n value that far exceeded the target (as determined by GPC; Table 1). Conversely, diethyl ether improved control over M_n and $Đ$, but overall monomer conversion was reduced. Polymerizations conducted in dichloromethane or toluene reached full conversion within 70 min, and the M_n values determined by GPC were typically in good agreement with the

Table 1. Optimization Conditions for the Polymerization of 2

entry	catalyst[a]	solvent	temperature (°C)	conversion (%)[b]	M_n (Da)[c]	target M_n (Da)[d]	$Đ$[e]
1	**4**	PhMe	25	100	39 210	30 000	1.16
2	**5**	PhMe	25	78	159 300	30 000	1.70
3	**6**	PhMe	25	100	15 490	30 000	1.26
4	**4**	PhMe	35	100	36 500	30 000	1.27
5	**4**	THF	25	70	95 400	30 000	1.56
6	**4**	DCM	25	100	35 300	30 000	1.29
7	**4**	ether	25	75	33 840	30 000	1.16
8[f]	**4**	MeCN	25	14	-	30 000	-

[a]See Scheme 1. [b]Determined after 70 min using ^{1}H NMR spectroscopy. [c]Determined by GPC. [d]Calculated using $[\mathbf{2}]_0$:$[\mathbf{Metal}]_0$ × 148.25. [e]Calculated using M_w/M_n. [f]No polymer material isolated for analysis.

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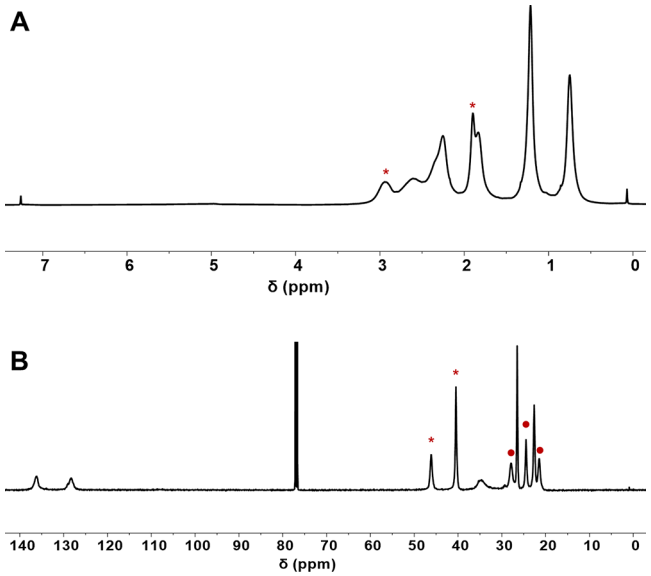


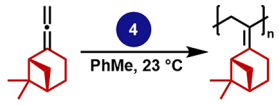
Figure 1. NMR spectra of **3** ($CDCl_3$). (A) ^{1}H NMR spectrum. Asterisks (*) represent bridgehead methine protons. (B) ^{13}C NMR spectrum. Asterisks (*) represent bridgehead carbons and circles (●) represent methylene carbons.

theoretical values (Table 1). Toluene was ultimately selected as solvent, however, as it afforded lower dispersity materials.

We also explored the polymerization of **2** under other conditions, but we note that vinyl-addition proved to be the most successful strategy. Cationic polymerization mediated by various Lewis acids (e.g., $AlCl_3$) proved ineffectual (or resulted in ring opening, as evidenced by two sets of vinyl resonances in the ^{1}H NMR spectrum; Table S2; Figure S27). Moreover, neither radical nor anionic polymerizations afforded any polymeric material (SI).

Using our chosen conditions, we successfully prepared materials exhibiting a wide range of molecular weights (M_n = 7504−99 580 Da; $Đ$ = 1.13−1.37) in excellent yields (87−99%; Table 2). We note that preparation of high M_n materials

Table 2. Representative Materials Accessed by Polymerization of 2



entry	M_n (Da)[a]	target M_n (Da)[b]	$Đ$[c]	yield (%)
1	7504	5040	1.37	87
2	18 920	12 750	1.19	90
3	26 420	25 050	1.13	89
4	52 030	49 960	1.23	88
5	66 270	74 570	1.23	94
6	99 580	102 000	1.31	99

[a]Determined by GPC. [b]Calculated using $[\mathbf{2}]_0$:$[\mathbf{Ni}]_0$ × 148.25. [c]Calculated using M_w/M_n.

(>60 000 Da) necessitated more dilute concentrations ($[\mathbf{2}]_0 \leq$ 0.10 M) in order to prevent gelation prior to full conversion. To assess the living character of these polymerizations, we monitored M_n as a function of monomer conversion (SI). The observed linear correlation, combined with a consistently low dispersity ($Đ$ < 1.30), provided strong evidence that the polymerization was living (Figure 2A). This conclusion was further corroborated through chain-extension experiments

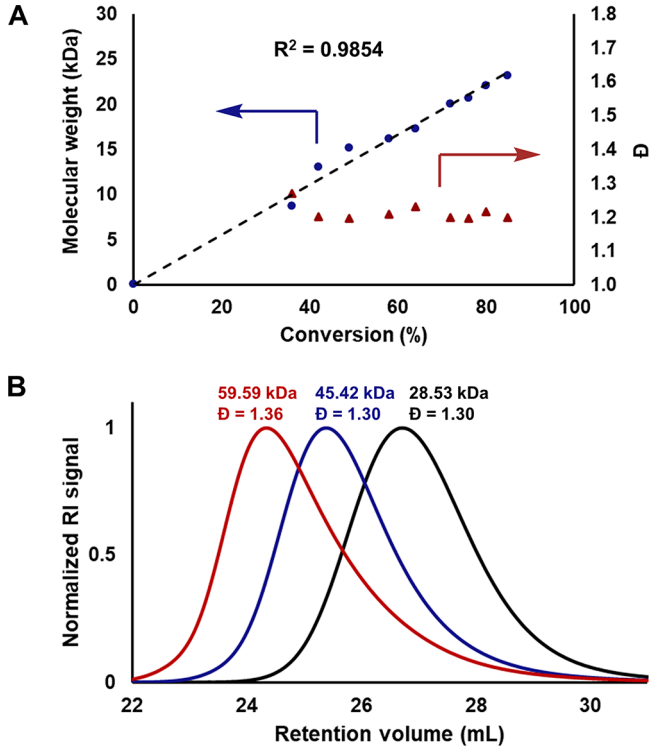


Figure 2. Determination of living character of the polymerization of **2**. (A) Plot showing molecular weight as a function of monomer conversion. (B) GPC traces and molecular weight data for chain extension experiments.

(Figure 2B). Briefly, the polymerization was monitored by ^{1}H NMR spectroscopy until complete consumption of **2** was observed. An aliquot of the reaction was removed and analyzed by GPC (following drying under high vacuum). A subsequent portion of **2** was then added to the reaction mixture. Upon consumption of the added monomer, an aliquot was removed and analyzed by GPC. This process was repeated an additional time, and GPC analyses of all three materials were consistent with fidelitous chain extension (Figure 2B). This living character starkly contrasts previous reports of pinene polymerizations, and the ability to chain extend **3** could facilitate the incorporation of pinane motifs into a broader range of soft materials.

Having synthesized our target materials, we sought to interrogate some of their bulk properties. For comparison, we also prepared a polymer from (1*S*)-(−)-*β*-pinene using standard[23] cationic conditions (**7**; M_n = 5812 Da; $Đ$ = 1.91; Scheme 1; SI). Thermogravimetric analysis of both materials revealed similar decomposition onset temperatures (cf. T_d = 294 °C for **3** versus 303 °C for **7**; Figures S6 and S8, respectively). Dynamic mechanical analysis (DMA) of powder samples revealed striking dissimilarities between **3** and **7**. While **3** exhibited a T_g = 160 °C (determined using $\tan(\delta)_{max}$; Figure S7), **7** exhibited a significantly lower T_g = 82 °C (which was consistent with previous reports;[23] Figure S9). As such, we expected **3** to exhibit some rigidification (i.e., crystallinity). To investigate, we performed wide-angle X-ray scattering (WAXS) on two different samples of **3** (M_n = 7504 and 79 910 Da). Consistent with our hypothesis, we observed two semicrystal-

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line peaks (average 2θ = 7.45° and 15.52°) in both samples (Figures S10 and S11). To ascertain if these data were associated with metastable phases, we annealed a sample of **3** (M_n = 66 090 Da) for 1 h under vacuum (SI). WAXS analysis of the annealed polymer was similar to the virgin material, albeit with larger average crystallite sizes (Figure S12; Table S3). In contrast, WAXS analysis of **7** revealed a peak at 2θ = 7.52° (which disappeared upon annealing; Figure S14) and another semicrystalline peak at 2θ = 15.74° (Figure S13). Moreover, the interchain *d*-spacing for annealed samples of **3** was smaller than that of annealed samples of **7** (cf. 11.46 versus 13.92 nm, respectively; Tables S3 and S4). Although preliminary, these data suggest that **3** exhibits tighter packing than **7**. Additional studies are currently underway to gain a deeper understanding of the morphologies associated with these polymers.

In conclusion, we have demonstrated that cumulation is a powerful strategy for accessing unique polymer architectures from biomass-derived monomers. The terpene congener, β-pinadiene, can be prepared in two steps on a multigram scale without the need for extensive purification. Vinyl-addition polymerization of this monomer is rapid at room temperature, exhibits living character, and affords high M_n materials with low dispersity. Importantly, these unique polymers retain their pinane core structure, and they exhibit physical properties that contrast those of poly(β-pinene). We believe this work presents a new opportunity to probe biomass-derived polymers and explore untapped chemical space in renewable polymer technology.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsmacrolett.0c00229.

Synthetic procedures, characterization, and additional experimental data (PDF)

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Author Contributions

This manuscript was written through contributions of all authors.

Notes

The authors declare no competing financial interest.

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